MINISTRY OF EDUCATION
IMAM ABDULRAHMAN BIN
FAISAL UNIVERSITY
COLLEGE OF COMPUTER SCIENCE
& INFORMATION TECHNOLOGY

وزارة التعليم
جامعة الإمام
عبدالرحمن بن فيصل
كلية علوم الحاسب
وتقنية المعلومات

جامعة الإمام عبدالرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

# Book Store Management System

## CSC236 – Object Oriented Programming 2 | Milestone 3
## Section 06 | Group (4)



| Student's Name | ID |
|---|---|
| Noor Murtada Alahmed (Leader) | 2230005190 |
| Almas Abdurahman Al-Qahtani | 2230002308 |
| Fatima Ahmed Al Shehab | 2230002074 |
| Nowra Faisal Alsalem | 2230007086 |
| Noor Mustafa Al-Hashim | 2230000582 |
| Maha Ahmed Aldhwaihi | 2230004763 |

# Table of Contents

***ER Diagram*** ..... ***3***

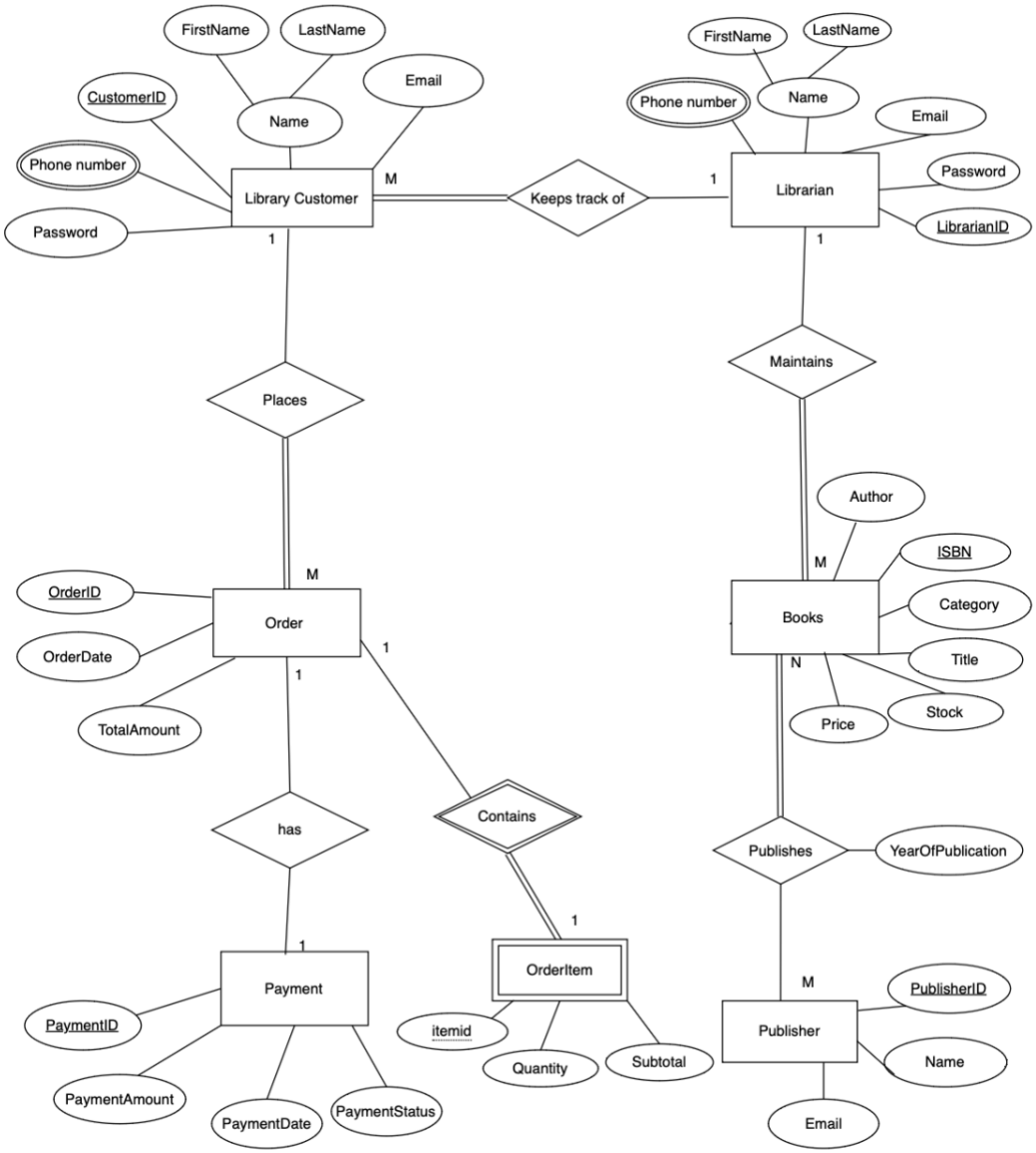
***Database Schema*** ..... ***4***

***SQL Tables*** ..... ***5***

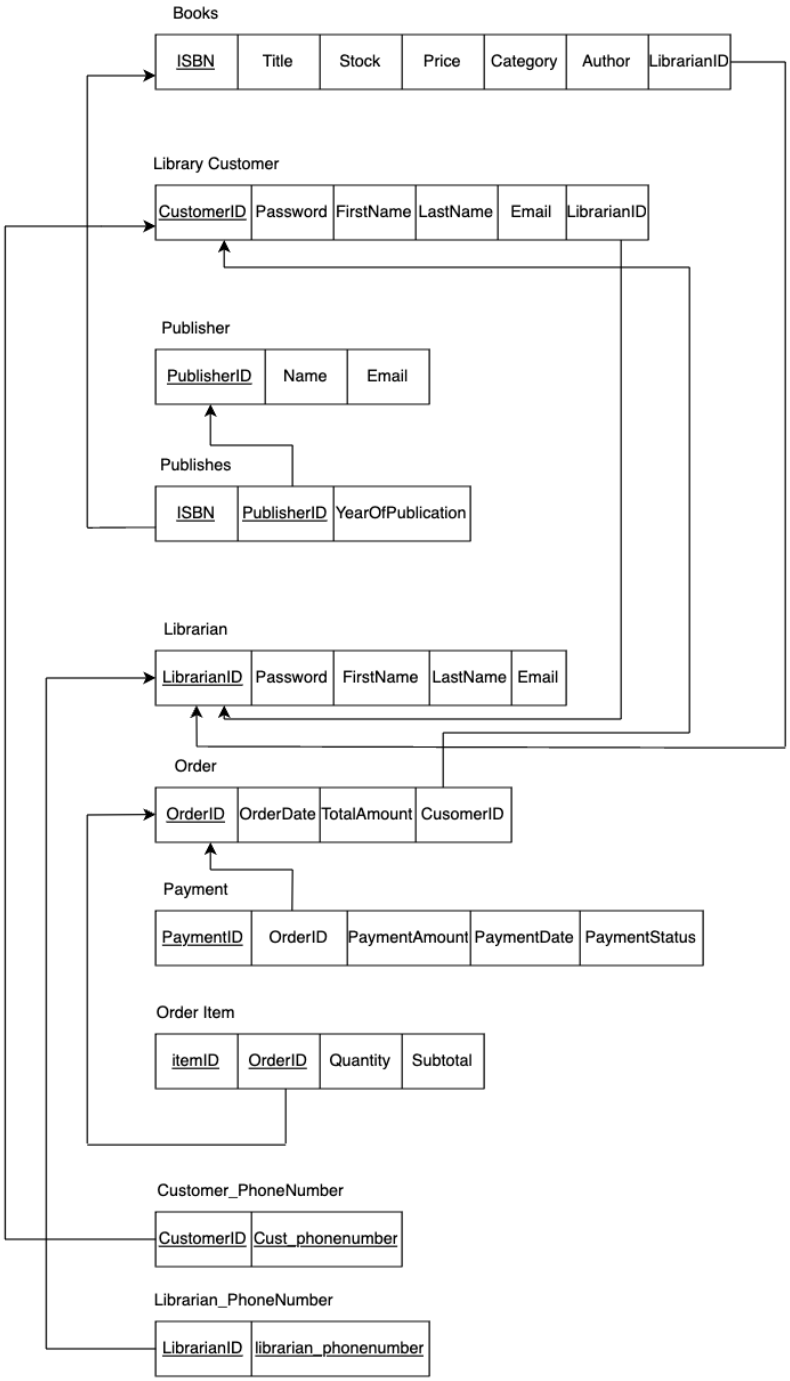
***Application Interface*** ..... ***7***

***Application Structure*** ..... ***16***

# ER Diagram


FirstName
LastName
Email
CustomerID
Name
Phone number
Library Customer
Password
M
1
Keeps track of
1
FirstName
LastName
Phone number
Name
Email
Librarian
Password
LibrarianID
1
Maintains
Places
Author
M
ISBN
M
OrderID
Order
Books
Category
OrderDate
1
1
N
Title
TotalAmount
Price
Stock
has
Contains
Publishes
YearOfPublication
1
1
OrderItem
Payment
M
PublisherID
PaymentID
itemid
Publisher
Quantity
Subtotal
Name
PaymentAmount
PaymentDate
PaymentStatus
Email

# Database Schema



# SQL Tables

## Customer table:

| CustomerID | FirstName | LastName | Email | PhoneNumber | Password |
|---|---|---|---|---|---|
| 100 | 2 | &Test Customer John | Test@example.com | 501234567 | 2 |
| 101 | John | Doe | johndoe@example.com | 501234567 | password123 |
| 102 | Jane | Smith | janesmith@example.com | 509876543 | password456 |
| 103 | Ahmed | Al-Qahtany | Qahtany@example.com | 505647382 | password789 |
| NULL | NULL | NULL | NULL | NULL | NULL |

## Librarian table:

| LibrarianID | FirstName | LastName | Email | PhoneNumber | Password |
|---|---|---|---|---|---|
| 200 | 1 | &Test Admin Noor | Test.murtada@gmail.com | 566864591 | 1 |
| 201 | Noor | Alahmed | noor.murtada@gmail.com | 566864591 | noor_123 |
| 202 | Huda | Al-Mansour | huda.mansour@library.sa | 512345678 | secureHuda99 |
| NULL | NULL | NULL | NULL | NULL | NULL |

## Books table:

| ISBN | Title | Author | Category | Price | Stock | YearOfPublication | PublisherID | LibrarianID |
|---|---|---|---|---|---|---|---|---|
| 1 | To Kill a Mockingbird | Harper Lee | Fiction | 40.00 | 10 | 1960 | 1 | 2001 |
| 2 | Pride and Prejudice | Jane Austen | Fiction | 30.00 | 5 | 1813 | 2 | 2001 |
| 3 | The Diary of a Young Girl | Anne Frank | Classic | 45.50 | 5 | 1947 | 3 | 2002 |
| 4 | Harry Potter and the Sorcerer's Stone | J.K. Rowling | Fantasy | 59.96 | 15 | 1997 | 4 | 2002 |
| 5 | Animal Farm | George Orwell | Classic | 35.75 | 10 | 1945 | 5 | 2001 |
| 6 | 1984 | George Orwell | Classic | 35.50 | 10 | 1949 | 5 | 2001 |
| 7 | The Little Prince | Antoine de Saint-Exupéry | Classic | 45.00 | 5 | 1943 | 6 | 2001 |
| 8 | The Great Gatsby | F. Scott Fitzgerald | Classic | 55.50 | 10 | 1925 | 7 | 2002 |
| 9 | The Catcher in the Rye | J.D. Salinger | Fiction | 40.00 | 5 | 1951 | 8 | 2002 |
| 10 | The Lord of the Rings | J.R.R. Tolkien | Fantasy | 80.00 | 15 | 1954 | 9 | 2002 |
| 11 | The Shining | Stephen King | Psychology | 14.99 | 35 | 1977 | 1 | 2001 |
| 12 | A Brief History of Time | Stephen Hawking | Psychology | 18.50 | 25 | 1988 | 1 | 2001 |
| 13 | The Hunger Games | Suzanne Collins | Psychology | 13.99 | 60 | 2008 | 1 | 2001 |
| 14 | The Picture of Dorian Gray | Oscar Wilde | Science ... | 9.99 | 20 | 1890 | 1 | 2001 |
| 15 | The Chronicles of Narnia | C.S. Lewis | Fantasy | 22.99 | 45 | 1950 | 1 | 2001 |
| 16 | The Art of War | Sun Tzu | Psychology | 7.99 | 75 | 500 | 1 | 2001 |
| 17 | The Martian | Andy Weir | Science ... | 16.99 | 50 | 2011 | 7 | 2002 |
| 18 | The Name of the Wind | Patrick Rothfuss | Fantasy | 17.50 | 40 | 2007 | 7 | 2002 |
| 19 | The Old Man and The Sea | Ernest Hemingway | Classic | 8.99 | 25 | 1952 | 7 | 2002 |
| 20 | The Power of Now | Eckhart Tolle | Thriller | 14.99 | 55 | 1997 | 7 | 2002 |
| 21 | The Girl with the Dragon Tattoo | Stieg Larsson | Thriller | 15.99 | 35 | 2005 | 7 | 2002 |
| 22 | The Road Less Traveled | M. Scott Peck | Psychology | 12.99 | 40 | 1978 | 7 | 2002 |
| 23 | The Stand | Stephen King | Science ... | 19.99 | 20 | 1978 | 9 | 2002 |
| 24 | The Immortal Life of Henrietta Lacks | Rebecca Skloot | Science ... | 16.99 | 35 | 2010 | 9 | 2002 |
| 25 | The Night Circus | Erin Morgenstern | Fantasy | 14.99 | 45 | 2011 | 9 | 2002 |
| 26 | The Lean Startup | Eric Ries | Classic | 18.99 | 50 | 2011 | 9 | 2002 |
| 27 | The Subtle Art of Not Giving | Mark Manson | Psychology | 15.99 | 70 | 2016 | 9 | 2002 |
| NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL | NULL |

-Publisher table:

| PublisherID | PublisherName | Email |
|---|---|---|
| 1 | HarperCollins | consumercare@harpercollins.com |
| 2 | Penguin Classics | info@penguinclassicpublishers.com |
| 3 | Bantam Books | info@bantambooks.com |
| 4 | Scholastic | info@scholastic.com |
| 5 | Signet Classics | info@signetclassics.com |
| 6 | Harvest Books | info@harvestbooks.com |
| 7 | Scribner | info@scribnerbooks.com |
| 8 | Little, Brown | info@littlebrown.co.uk |
| 9 | Mariner Books | info@marinerbooks.com |
| NULL | NULL | NULL |

-OrderItem table:

| orderID | ISBN | quantity | subTotal |
|---|---|---|---|
| 100000 | 1 | 1 | 40.00 |
| 100000 | 2 | 1 | 30.00 |
| 100001 | 3 | 2 | 40.00 |
| 100002 | 4 | 4 | 35.75 |
| NULL | NULL | NULL | NULL |

-Order table:

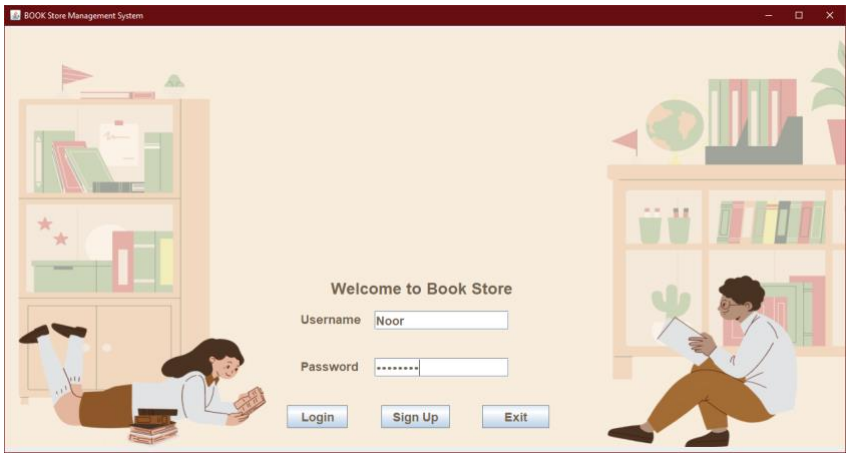
| orderID | orderDate | totalAmount | userID |
|---|---|---|---|
| 100000 | 2025-04-01 | 70.00 | 100 |
| 100001 | 2025-04-02 | 80.00 | 101 |
| 100002 | 2025-04-03 | 35.75 | 102 |
| NULL | NULL | NULL | NULL |

-Payment table:

| paymentID | paymentDate | paymentAmount | paymentStatus | orderID | userID |
|---|---|---|---|---|---|
| 200000 | 2025-04-01 | 70.00 | successful | 100000 | 100 |
| 200001 | 2025-04-02 | 80.00 | faild | 100001 | 101 |
| 200002 | 2025-04-03 | 35.75 | successful | 100002 | 102 |
| NULL | NULL | NULL | NULL | NULL | NULL |

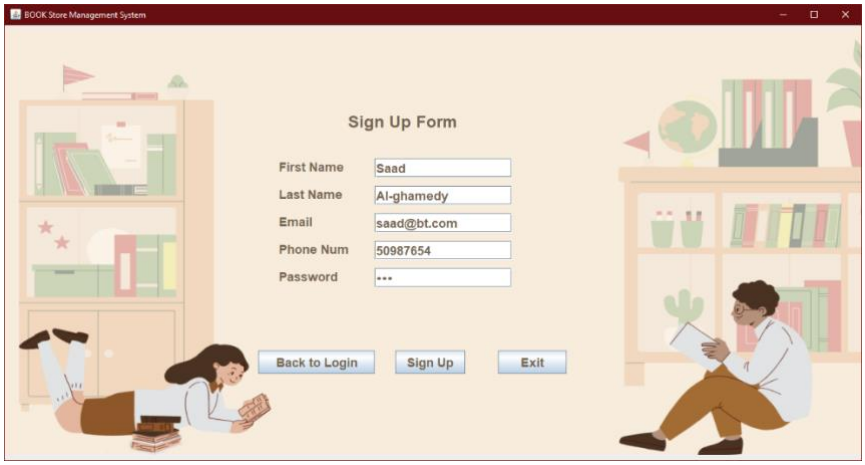
# Application Interface

## Log-in page:



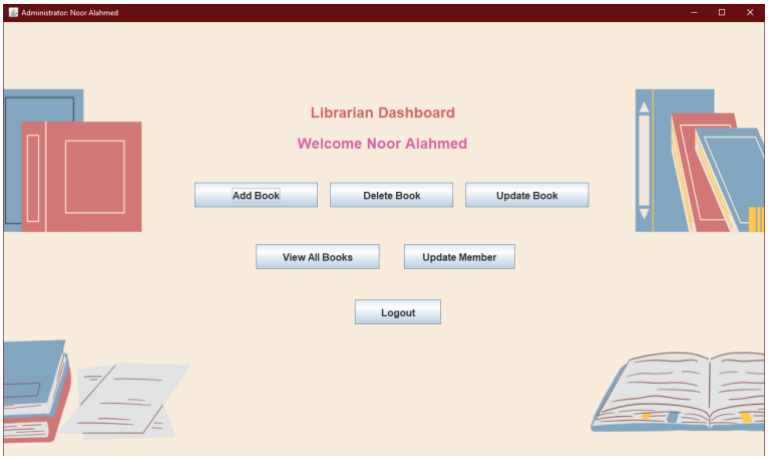

The uploaded interface represents the Login Page for the eBook Store Management System. The user can enter their Username and Password to log in. There are three buttons: Login to enter the system, Sign Up to create a new account, and Exit to close the application.

## Sign-up page:



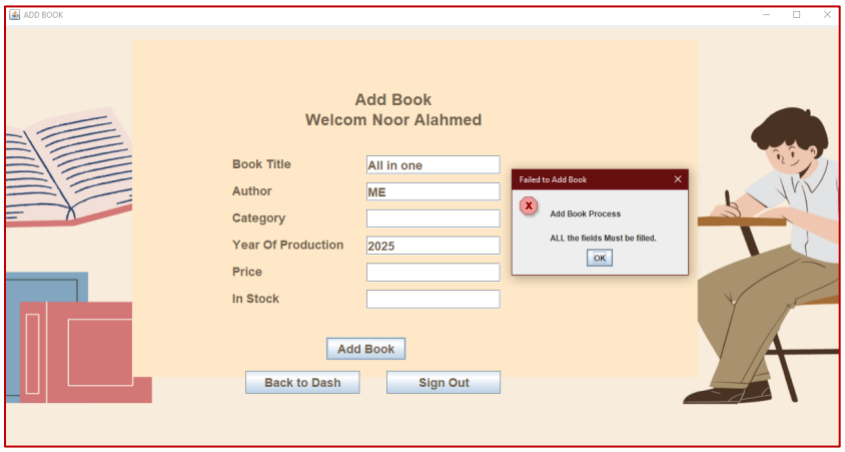

The uploaded interface represents a Sign-Up Form for the eBook Store Management System. It allows users to enter their personal information, including their first name, last name, email address, phone number, and password. The form features three buttons: "Back to Login," which takes users back to the login page; "Sign Up," which submits the information to create a new account; and "Exit," which closes the application.

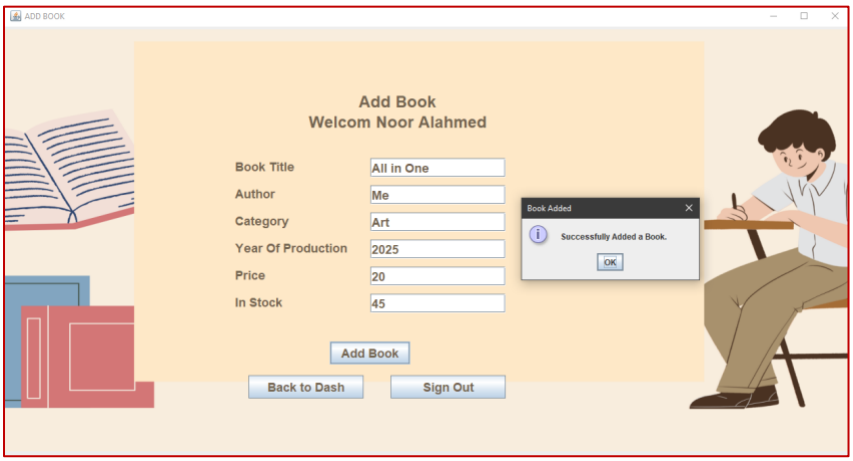
Librarian Dashboard page:




This is the Librarian (Admin) Dashboard of the Library Management System, designed to help the librarian efficiently manage the library's resources and users. Through this dashboard, the librarian can add new books, delete or update existing book records, and view all books currently available in the system. It also allows access to the list of registered library members. Additionally, there are features for searching for specific books or members and editing existing records. The dashboard includes a logout option to securely exit the admin session.

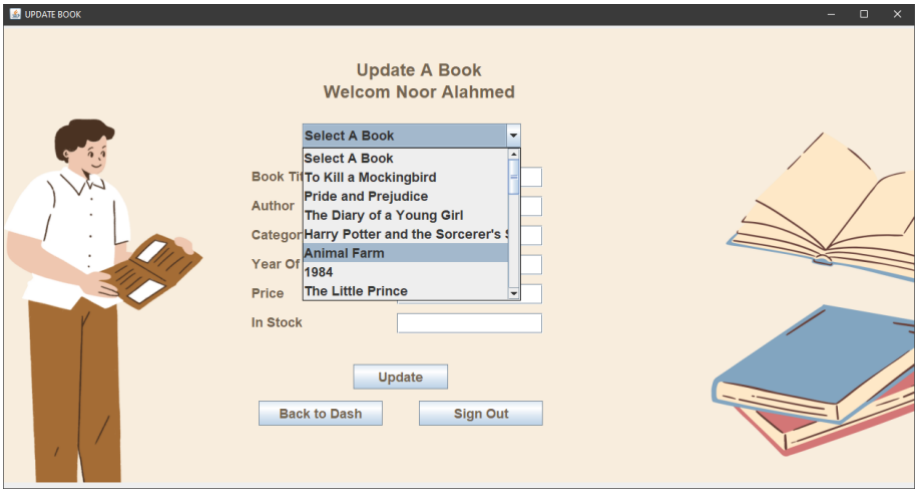
Librarian (adding a book) page:





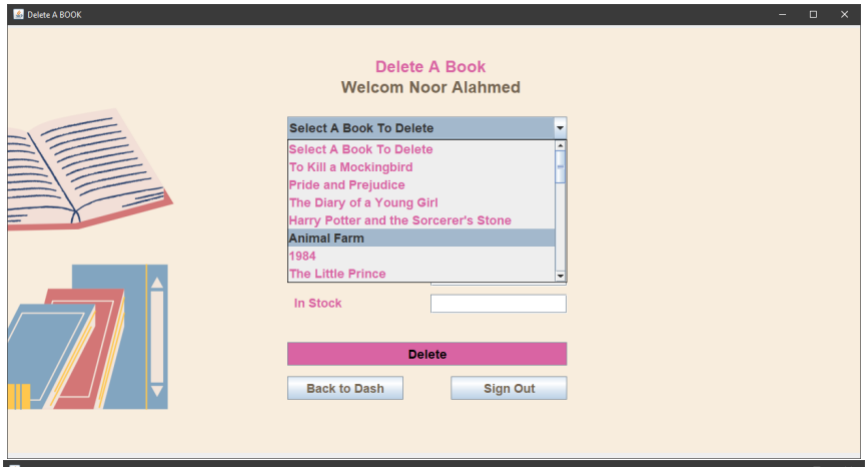

This page is an admin interface for adding new books to the library management system. It includes a form with fields for the book's Title, Author, Category, and Year of Publication. After filling in the details, the admin can click the "ADD" button to submit the book information to the system, and if the admin did not fill all the fields an error message shows up “all the fields must be filled up”, if adding a book was successful, a message shows up “Successfully Added a Book”.

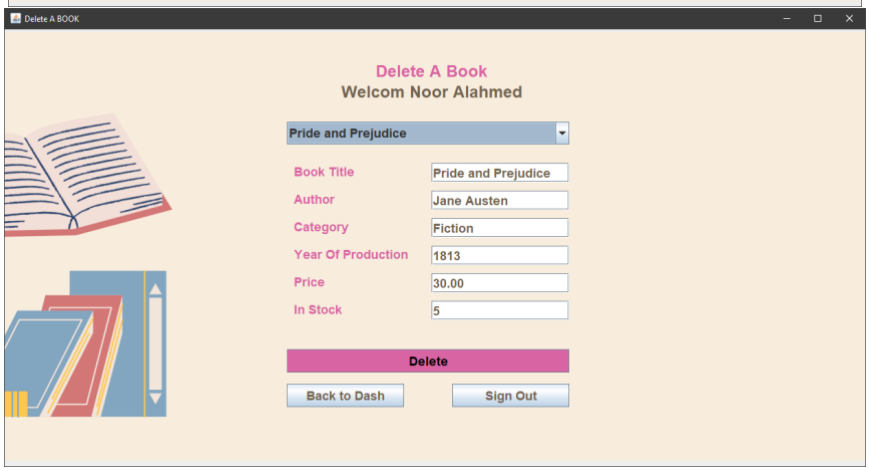
Librarian (updating a book) page:




The uploaded interface is for updating a book in the eBook Store Management System. The user, Noor Alahmed, can select a book from a dropdown menu, update details like author, category, year, price, and stock availability. There are buttons to Update the changes, Back to Dash for returning to the dashboard, and Sign Out to log out. The background features an illustration of books.

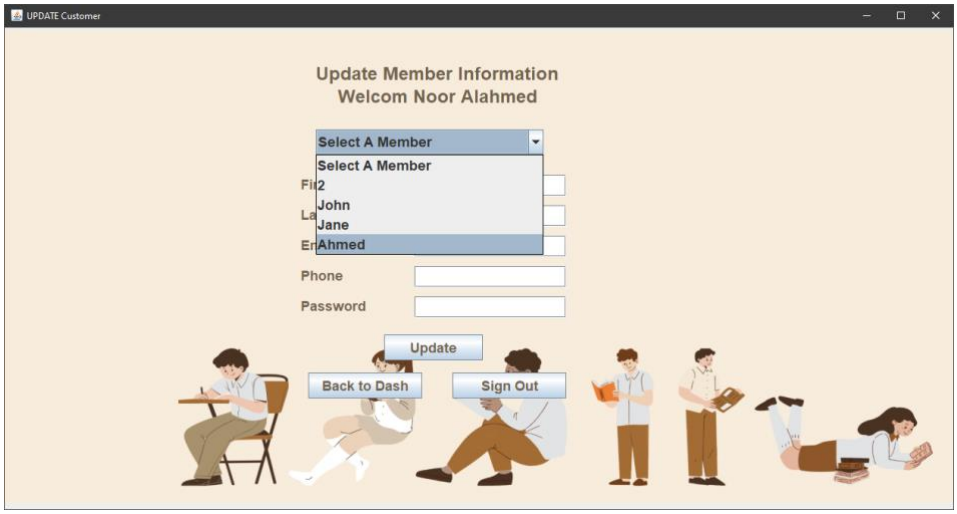
Librarian (Deleting a book) page:



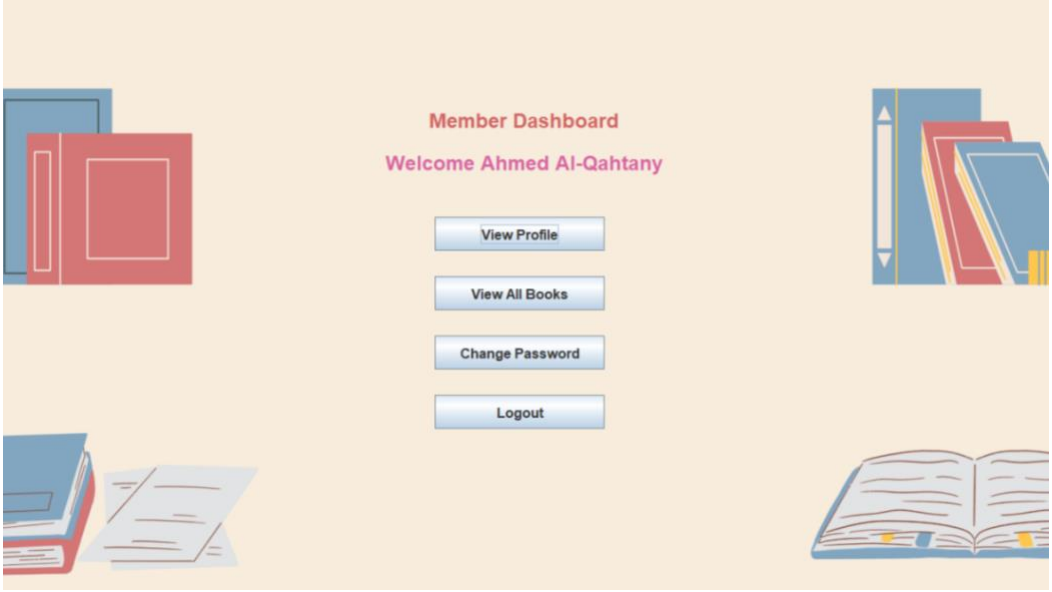

The uploaded interface is for the Delete a Book page in the eBook Store Management System. The admin can select a book from the dropdown menu to delete, with options like *To Kill a Mockingbird*, *Animal Farm*, and *1984*. The user can also mark the stock availability. There are buttons for Delete to remove the selected book, Back to Dash to return to the dashboard, and Sign Out to log out. The background features illustrations of books, continuing the theme of the system.

Librarian (Updating Member) page:




This page is a member information update interface for a management system. It greets the logged-in user and allows them to select a member from a dropdown list to update their details, including name, email, phone, and password. Users can then choose to update the information, return to the dashboard, or sign out.

Member dashboard page:

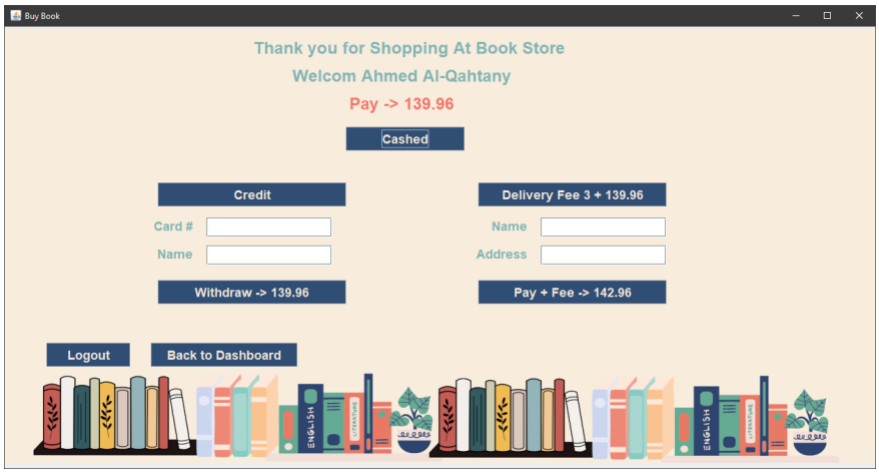


This is the Member Dashboard of the Library Management System, designed to provide library users with easy access to their personal information and book-related activities. From this interface, members can view their profile details, browse the complete list of books available in the library, and check the books they have currently borrowed. There is also an option to change their account password for security purposes. The dashboard includes a logout button to securely end the session.

Member & Admin (View all books, add to basket, buy, and borrow) page:

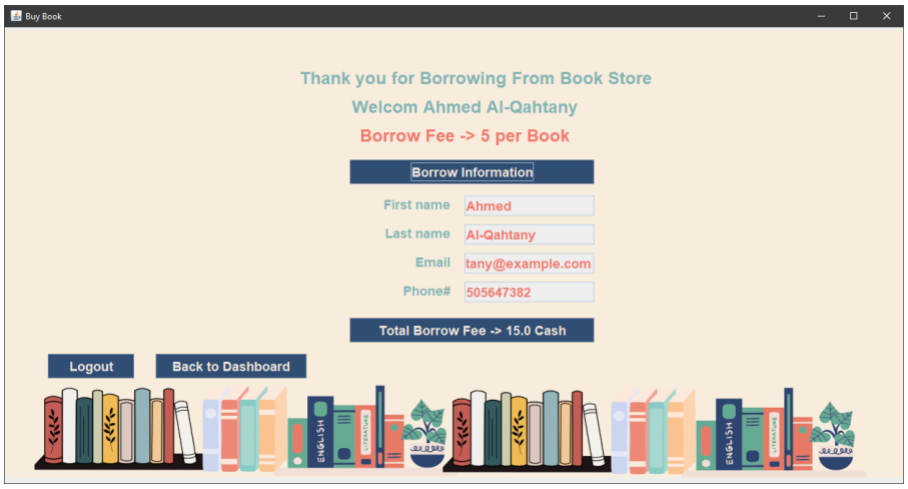
The uploaded interface is for the View Books Information page, showing details like title, author, category, price, and stock availability of the selected book. The user can add books to the basket, view the total price, and choose to Buy or Borrow. The page also includes options to Logout and Back, with a book-themed design at the bottom.

Member (payment) page:




The uploaded interface is the Payment page, showing the total payment of 139.96. The user, Ahmed Al-Qahtany, can pay via Cash or Credit, entering card details for the latter. A Delivery Fee of 3 is added, making the total 142.96. Options to Logout or Back to Dashboard are also available

Member (Borrowing) page:



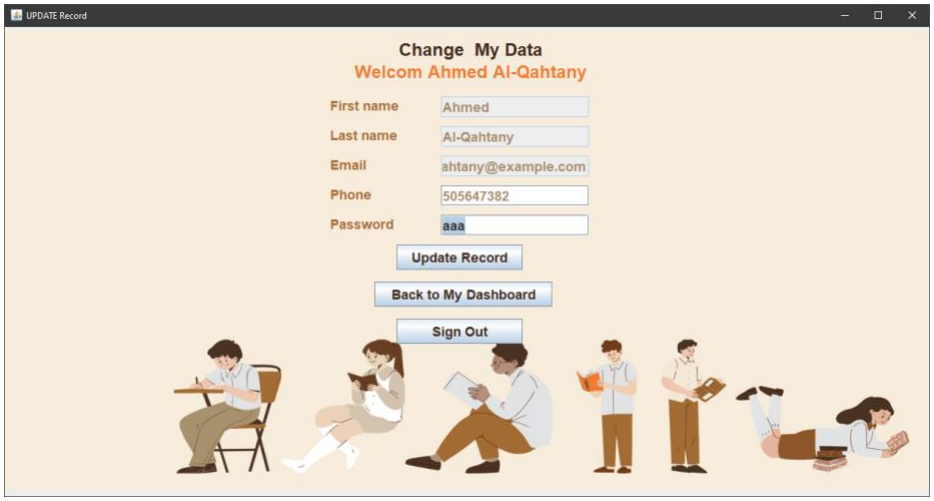

This page is a book borrowing confirmation interface for a bookstore system. It greets the user and displays a borrowing fee of 5 per book. The form shows the user's personal information (first name, last name, email, and phone number), and calculates the total borrow fee—in this case,

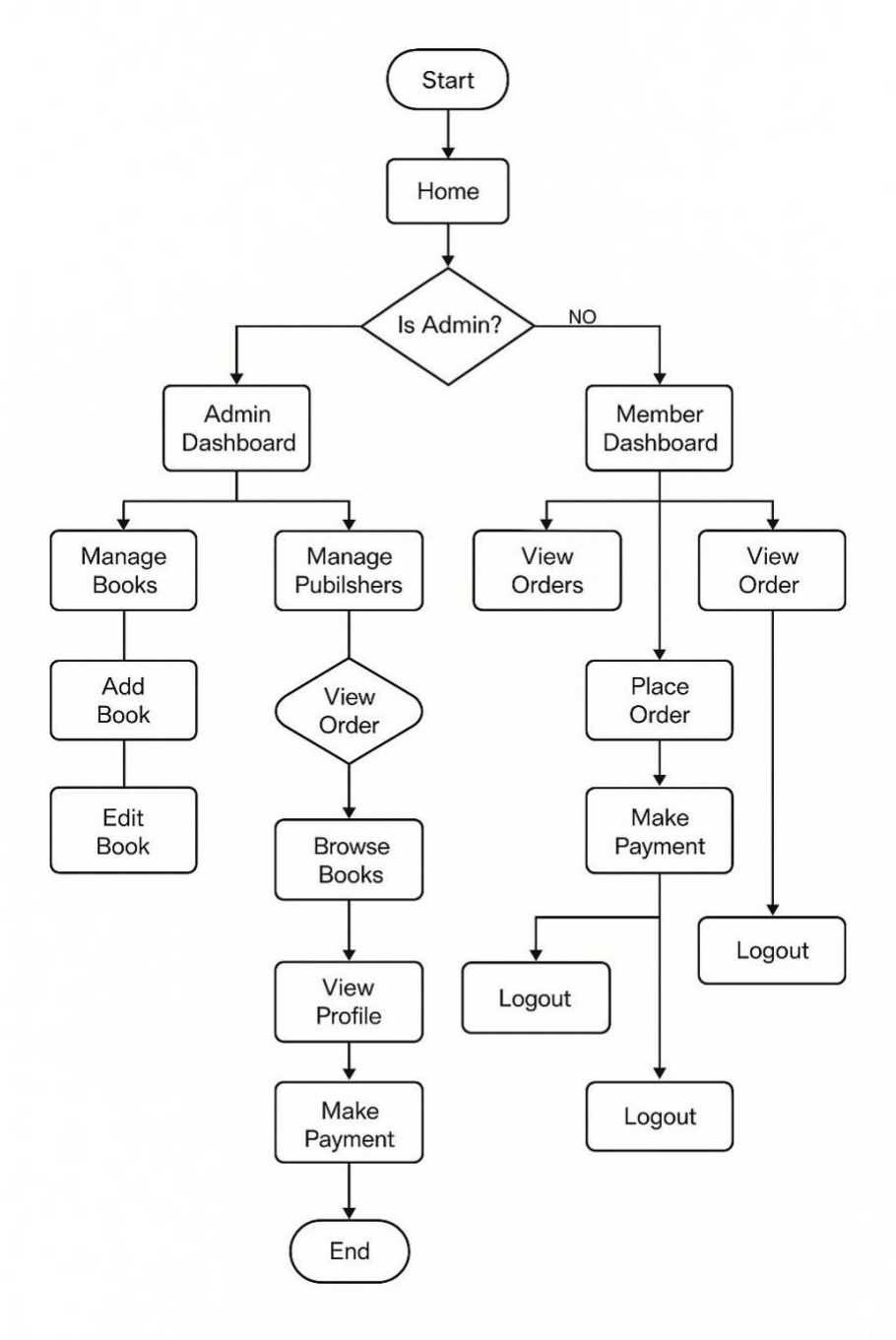
15.0 cash, suggesting that three books were borrowed. The page includes buttons to either log out or return to the dashboard.

View Profile page:

This page is a user profile interface where the logged-in user, can view and update their personal information. It includes editable fields for first name, last name, email, phone number, and password. The user can navigate back to the dashboard or sign out using the provided buttons.

Member (Change Password) page:



The "Change My Data" page allows users like Ahmed Al-Qahtany to view and update their password. The "Update Record" button saves changes, while links to My Dashboard and Sign Out provide quick navigation.

# Application Structure